

MATHEUS AVELINO DE JESUS

Full Stack Software Engineer · .NET & C# · Python · JavaScript & TypeScript · Angular & React · APIs & Microservices · Messaging · AWS & Azure · Software Architecture

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PROFESSIONAL SUMMARY

Software engineer with over **8 years** of full stack experience, focused on **.NET and C#** back ends and also working across Python, JavaScript, TypeScript and Node.js. Since 2022 in the financial sector, building and supporting critical integrations for credit unions — Brazil's Central Bank (Bacen), Pix, digital signature and the EFD-Reinf tax obligation — in environments that mix monoliths and microservices.

I work with REST/RESTful APIs in ASP.NET Core, asynchronous communication with Apache Kafka and RabbitMQ, and data modelling and tuning on SQL Server, Oracle (PL/SQL), PostgreSQL, MySQL and MongoDB, plus services on AWS and Azure. I apply Domain-Driven Design, SOLID, Clean Code, Design Patterns and microservice architecture to reduce coupling and keep systems maintainable. Strong in production: critical incident diagnosis, performance bottleneck analysis, batch parallelisation and monitoring with Datadog. On the front end I deliver interfaces with Angular, React, Vue, TypeScript and hybrid apps in Ionic — with unit tests, Swagger/OpenAPI documentation, Git, Docker and CI/CD, inside agile squads (Scrum).

KEY STRENGTHS

Regulated financial integrations — Bacen, Pix, EFD-Reinf and digital signature in credit-union systems serving over 1.7 million members: a domain demanding compliance, traceability and availability.

Batch processing performance — Restructured routines consuming multiple endpoints sequentially into parallel processing (Parallel.ForEach and Task), with a significant reduction in response time.

Solid back end on a real full-stack base — End to end: data modelling, APIs, messaging and interfaces in Angular, React and Vue.

Sectors — Banking & finance · Credit unions · Payments (Pix) · Regulatory integrations (Bacen, EFD-Reinf) · Insurance (corporate life and agricultural) · Healthcare · E-commerce · HR and corporate administrative systems.

PROFESSIONAL EXPERIENCE

Back-End Developer Mar 2022 — present
Mouts TI

Technology consultancy in the financial sector. Projects for the Ailos System — 14 affiliated credit unions and over 1.7 million members across 8 states — and the CrediSIS System, with 14 credit unions and 80+ service points. Regulated environment with defined processing windows and audit-trail requirements.

- Developed and evolved integration services with the **Central Bank (Bacen)**, digital signature platforms and the **Pix** payment scheme, which runs 24x7 with roughly 10-second settlement and allows no scheduled downtime.
- Implemented the **EFN-Reinf** tax obligation, transmitted to SPED with a digital certificate and the tax authority's layout, due on the 15th of the following month — late filing incurs fines of R\$500 to R\$1,500 per period.
- Diagnosed a performance bottleneck in routines consuming multiple endpoints sequentially and restructured the flow with **parallel processing** (Parallel.ForEach and Task), cutting batch load times — decisive because these routines run in night windows with hard deadlines, where overrun means reprocessing and regulatory risk.
- Implemented dynamic handling of generic payloads with **Reflection**, simplifying serialisation of variable-contract responses, removing duplicated code and lowering maintenance effort per new integration.
- Structured asynchronous communication between services with **Apache Kafka** and **RabbitMQ**, decoupling long-running processes from the synchronous API flow so queues absorb third-party instability without losing messages.
- Designed and implemented REST/RESTful APIs in **ASP.NET Core** applying DDD, SOLID and Design Patterns, in a hybrid scenario where new microservices coexist with the legacy monolithic core.
- Modelled and tuned data structures in SQL Server and stored procedures in PL/SQL, reviewing execution plans, queries and indexes to sustain transaction volumes concentrated around closing and due dates.
- Monitored production applications with **Datadog**, tracking metrics and logs to catch degradation before downtime and leading root-cause analysis of critical failures.
- Standardised API documentation with Swagger/OpenAPI and XML comments and expanded unit-test coverage, reducing regression risk in a restricted deploy window. Published and configured applications on IIS, used AWS services for hosting, storage and scheduled routines, and built AI proofs of concept in Python.

Tech: C#, .NET Core, ASP.NET Core MVC and Web API, Entity Framework Core, Dapper, LINQ, SQL Server, Oracle/PL-SQL, Apache Kafka, RabbitMQ, AWS, Datadog, Docker, IIS, Swagger/OpenAPI, xUnit, Python, JavaScript, Node.js, React, Git, Scrum.

Full Stack Developer

Jun 2021 — Mar 2022

GCOM

Digital solutions company with clients in e-commerce. Work on new products and support of platforms in operation.

- Worked on new products and support of e-commerce platforms, across monolithic applications and microservices hosted on **Azure**.
- Modelled and implemented non-relational persistence in **MongoDB** for read-intensive scenarios and flexible catalogue structures, complementing the relational SQL Server base.
- Built and consumed REST/RESTful APIs exposed through an **API Gateway**, integrating internal systems and third-party services and reducing coupling between platform domains — a requirement for sustaining e-commerce seasonal peaks.
- Took part in developing a hybrid mobile application in **Ionic** integrated with the internal system, extending business operations beyond the desktop.
- Implemented integrations with **TOTVS RM Chronus** (HR) and the **Fluig** platform, automating administrative flows previously executed manually.
- Maintained and evolved legacy **VB6** systems on the Sisbase database and procedures, keeping operations stable while new modules migrated to .NET in an incremental modernisation where the legacy still generates revenue.
- Developed web interfaces with **Angular, React, Vue and TypeScript**, standardising components, and supported the rollout of observability across cloud services.

Tech: C#, .NET Core MVC, Entity Framework Core, SQL Server, MongoDB, PL/SQL, VB6, Sisbase, Angular, React, Vue, TypeScript, JavaScript, jQuery, Ajax, HTML5, CSS3, Bootstrap, Ionic Framework, Azure, AWS, API Gateway, Git, Azure DevOps, Postman, Swagger.

Full Stack Developer

Mar 2021 — Jun 2021

Tria Software

Software house working in the hospital sector. Project for Palidel, a US client based in California, with a critical one-month delivery deadline.

- Took over the front end of the hospital solution under a critical **30-day** deadline, sequencing deliveries by functional priority to meet the agreed schedule.
- Developed the hybrid application in **Ionic Framework** with native plugins, enabling push notifications and offline storage — essential in a hospital environment with unstable connectivity.
- Built interfaces in **Angular 11**, HTML5, CSS3 and Bootstrap 4 focused on usability and responsiveness, integrating them with .NET Core MVC services and the SQL Server database through Entity Framework Core to keep the mobile app and central system consistent.

Tech: Angular 11, Ionic Framework (Native Plugins), TypeScript, JavaScript, HTML5, CSS3, Bootstrap 4, C#, .NET Core MVC, Entity Framework Core, SQL Server.

Full Stack Developer

Jan 2020 — Mar 2021

Adobe Crefisa

Finance and credit sector. Squad dedicated to evolving and stabilising internal corporate systems.

- Joined the squad responsible for continuous improvement of internal systems, from problem discovery with business areas through to production delivery.
- Diagnosed and fixed recurring defects in live systems, prioritising **root-cause analysis** over point fixes, which cut ticket reopenings and freed squad capacity for new features.
- Developed new features in REST/RESTful APIs with **.NET Core Web API**, applying Domain-Driven Design to separate business rules from infrastructure.
- Built hybrid mobile applications in **Ionic** using native plugins (Camera, Geolocation and Storage) to enable field operations directly from a phone.
- Implemented interfaces in **Angular 7+** integrated with internal APIs, and delivered integrations with the **TOTVS RM** ERP plus SQL Server modelling with Entity Framework Core.

Tech: C# 7, .NET Core 2.2, ASP.NET Core Web API, Entity Framework Core, DDD, SQL Server, Angular 7+, Ionic Framework, TypeScript, JavaScript, TOTVS RM, Git (Bitbucket and Azure DevOps), Postman.

Full Stack Developer (contractor)

Jan 2018 — Jan 2020

Berkley Seguros & Ezze Seguros

Large insurance companies. Digital quotation products and support of corporate systems in the insurance sector.

- Developed a **corporate life insurance quotation system** for Berkley, translating pricing rules and acceptance criteria into reusable services, which reduced the sales team's dependence on manual quotes.
- Drove the completion of an **insurance portal** incorporating the agricultural product's quotation engine, delivering the solution to production.
- Architected the applications following **Domain-Driven Design**, isolating the insurance domain from presentation and persistence layers and simplifying the addition of new products.

- Implemented REST/RESTful APIs in .NET Core Web API and applications in .NET Core MVC, with contract documentation and validation via **Swagger** and unit-test coverage.
- Developed **scheduled jobs** in .NET Framework to process policy and proposal data outside peak hours, isolating bulk processing from the transactional applications brokers use during business hours.
- Ran production support, system integrations and defect fixing, with IIS deployments and versioning in Git, Azure DevOps and TFS.

Tech: C# 8, .NET Core 3.0 and 3.1, .NET Framework, ASP.NET Core Web API and MVC, Entity Framework Core, DDD, SQL Server, AngularJS, JavaScript, HTML5, CSS3, Bootstrap 4, Swagger, IIS, Git, Azure DevOps, TFS.

TECHNICAL SKILLS

Languages: C#, Python, JavaScript, TypeScript, SQL, PL/SQL, VB6, HTML5, CSS3

Back end: .NET / .NET Core, ASP.NET Core MVC and Web API, Entity Framework Core, Dapper, NHibernate, LINQ, Node.js, Hangfire, IdentityServer4, REST APIs, async and parallel processing, Reflection, distributed services

Architecture & practices: Microservices, DDD, SOLID, Clean Code, MVC, Design Patterns, Repository Pattern, Dependency Injection, BDD, event-driven architecture, domain modelling

Cloud: AWS (S3, EC2, Lambda) and Azure (Blob Storage, Cloud Functions, Cloud Storage, microservice hosting)

Testing & quality: Unit and integration tests, Postman, Swagger/OpenAPI, code review, refactoring, technical documentation

Corporate platforms: TOTVS RM, TOTVS RM Chronus (HR), Fluig

Front end: Angular, React, Vue.js, TypeScript, JavaScript, jQuery, Ajax, HTML5, CSS3, Bootstrap, Ionic Framework

Databases: SQL Server, Oracle, PostgreSQL, MySQL, MongoDB, Sisbase; stored procedures, views, indexes, modelling, normalisation and query tuning

Messaging: Apache Kafka, RabbitMQ, async processing, queues and topics, service decoupling

DevOps & versioning: Docker, Git, Azure DevOps, Bitbucket, Jenkins, Bamboo, TFS, SVN, CI/CD, pipelines, containerisation, IIS deployments

Observability: Datadog, structured logging, production monitoring, incident analysis and failure diagnosis

Methodologies: Scrum, multidisciplinary squads, agile ceremonies, technical refinement

EDUCATION

Systems Analysis and Development — Technologist Degree

Centro Universitário das Faculdades das Américas (FAM)

Completed: July 2022

High School

E.E. Prof.^a Araci Zebral Teixeira

Completed: December 2016

ADDITIONAL TRAINING

.NET Software Architecture · REST with ASP.NET Core Web API · Mastering ASP.NET MVC Core · SPA Development with Angular

desenvolvedor.io — Eduardo Pires

January to April 2020 · completed

Advanced C# · Angular · HTML and CSS

DevMedia

March–April 2018 · completed

LANGUAGES & ADDITIONAL INFORMATION

Languages: Portuguese (native); English with fluent technical reading and conversation in development.

Location and availability: São Paulo – SP, Brazil. Immediate start. Active CNPJ, with flexible contracting models.